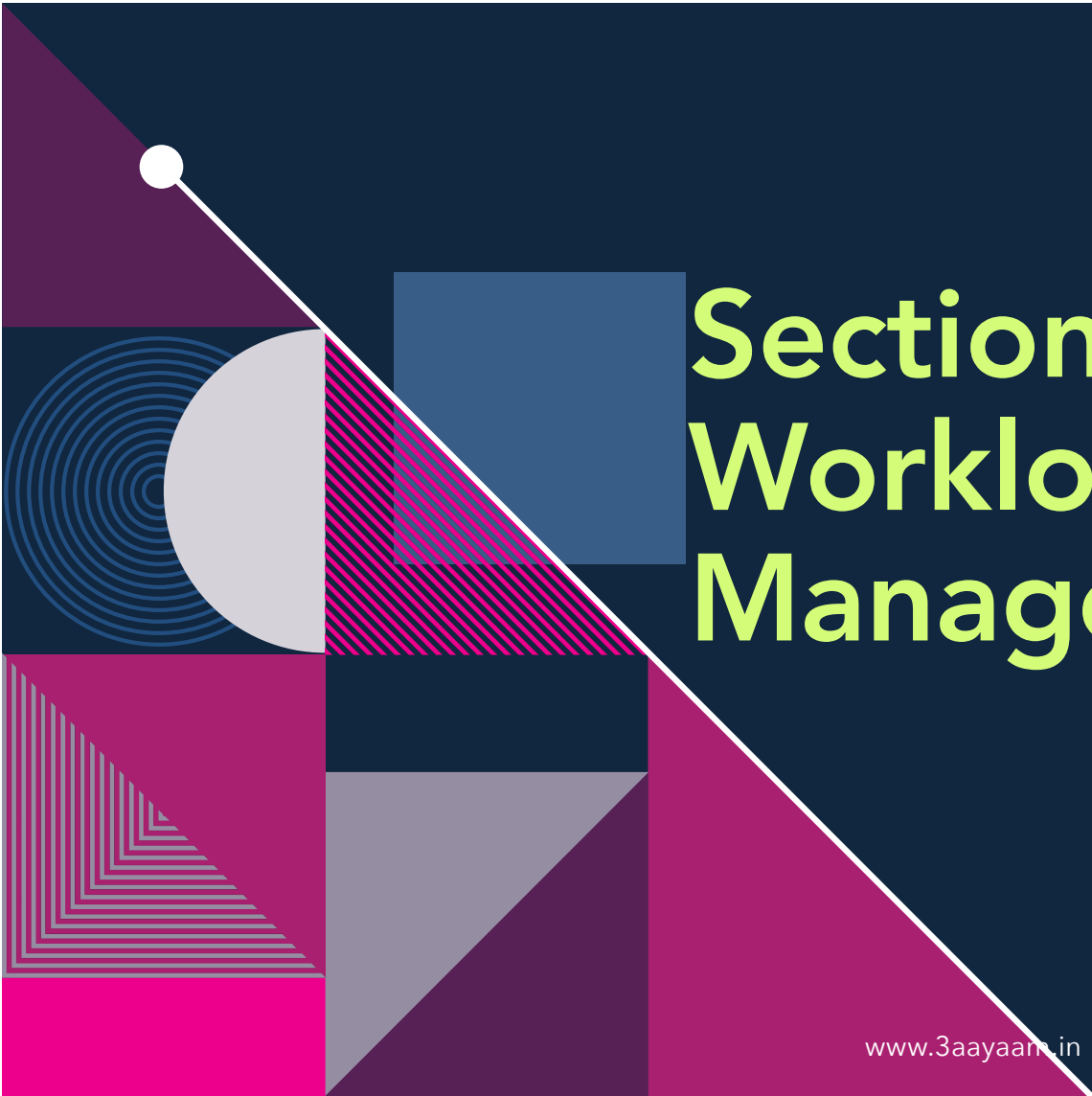




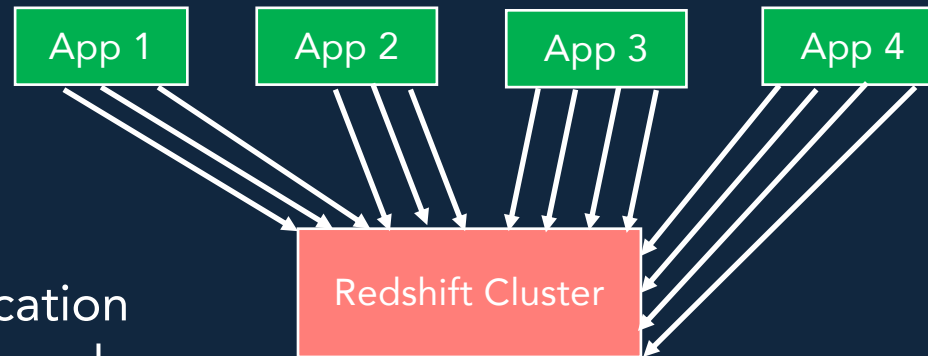
Section 5 - Workload Management (WLM)

Agenda

- WLM Primer
- Query Queue
- Concurrency Scaling
- Short Query Acceleration
- Query Assignment
- Query Monitoring Rule
- Automatic and Manual WLM

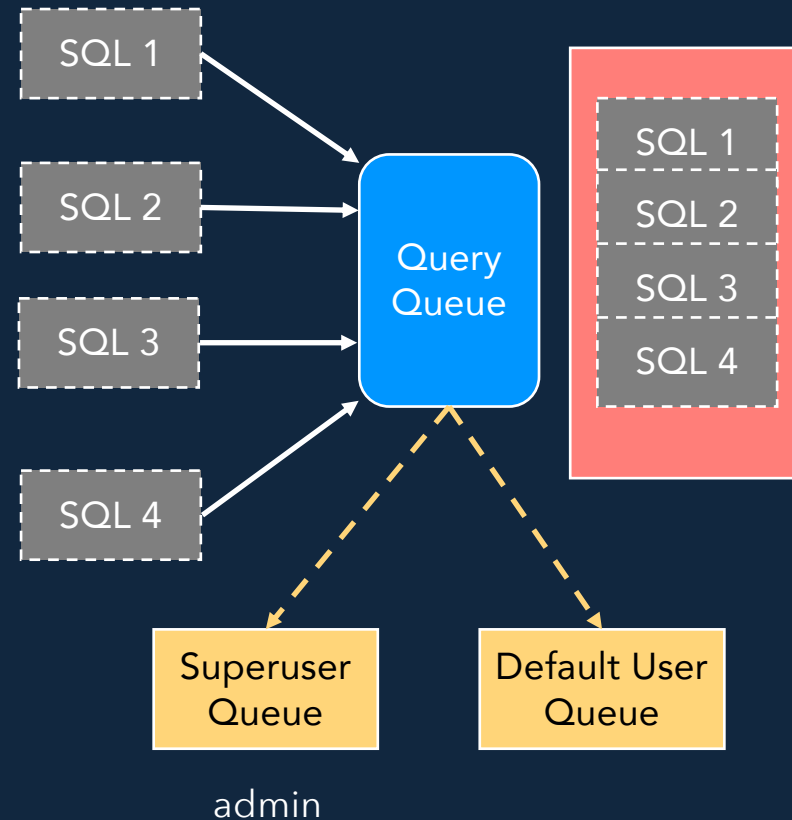
Why WLM?

- Concurrent queries.
- Resource utilization (CN).
- How to prioritize?
- WLM controls resource allocation to different types of queries and users.
- Define query queues, user groups.
- WLM
 - Automatic WLM
 - Manual WLM



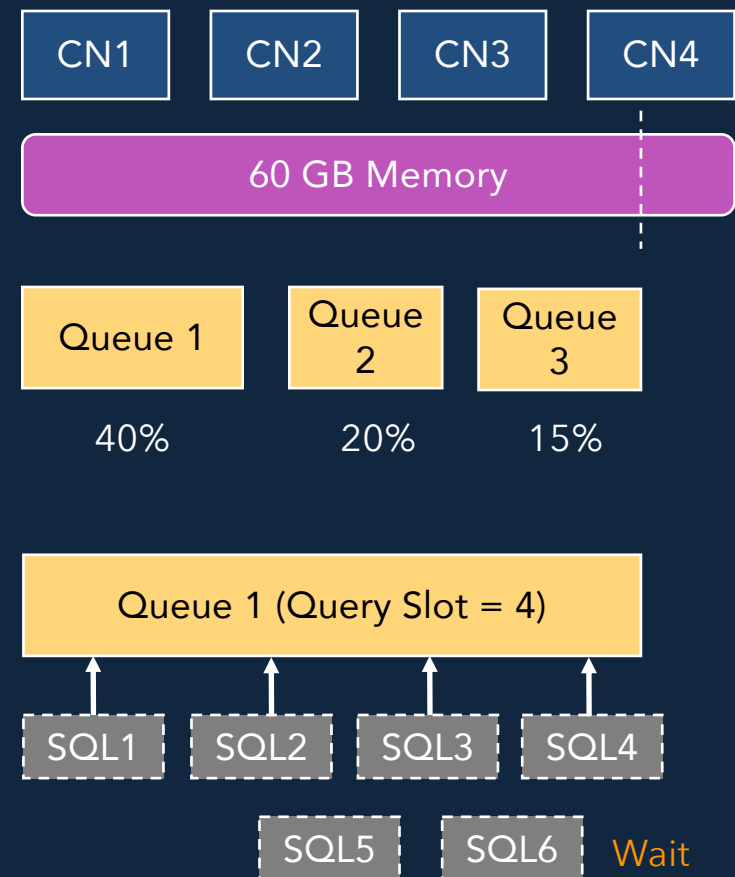
WLM Primer

- Query queues?
- What if ...
 - SQL 1 takes 45 mins.
 - SQL 2 takes 5 sec.
 - SQL 3 is the most critical one.
- Default queues
 - Superuser queue
 - Default user queue
 - 5 queries concurrently
- Define additional query queues using WLM



Query Queue

- Contains certain amount of configurable memory of the cluster (CN).
- Concurrency Level (Query Slot) per queue
 - Each SQL will get 10% of the cluster's memory.
 - Default concurrency level is 5.
- **Max slot count** across all defined queues is **50**.
- Queue Assignment
 - User Groups
 - Query Groups

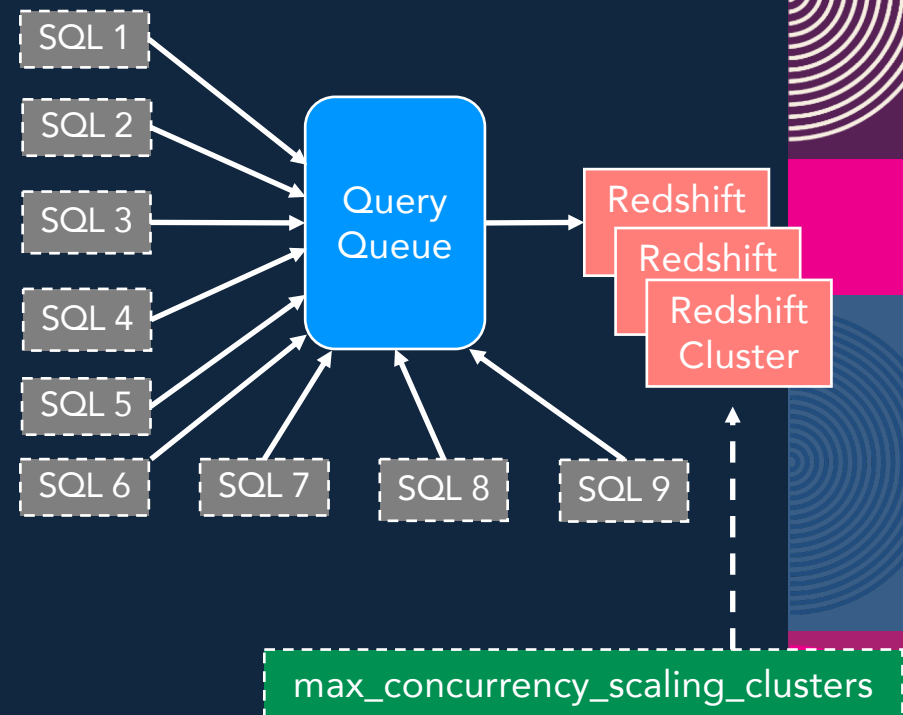


Query Queue

- Memory percent to use.
 - Each queue gets certain amount of memory of the cluster (CNs).
 - Available memory of a queue is equally divided among the Query Slots.
- Concurrency Scaling
- Query Monitoring Rules

Concurrency Scaling

- Thousands of concurrent users and queries instead of hundreds.
- Adds more cluster capacity.
- Supports both reads and writes.
- Supported write operations.
 - INSERT, DELETE, UPDATE
 - COPY, CTAS
 - MV refresh (without aggregations)
 - DDL operations are NOT supported.
 - Not supported for tables with DISTSTYLE ALL.
- **SVCS_CONCURRENCY_SCALING_USAGE**



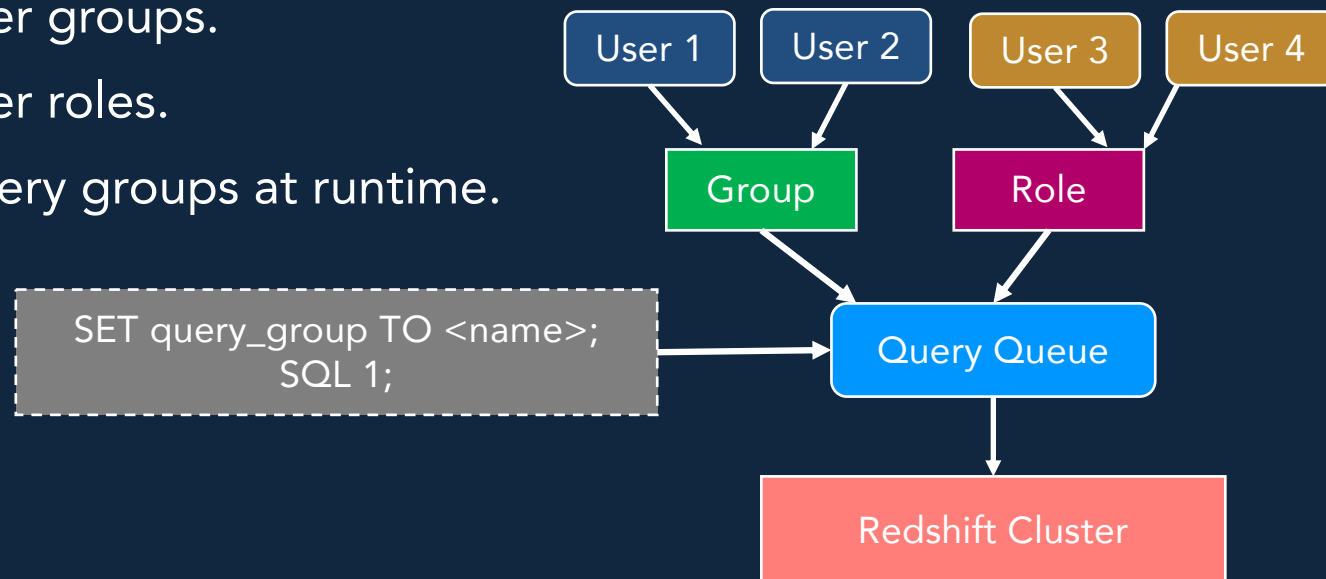
Short Query Acceleration (SQA)

- Prioritize short running queries ahead of long running ones.
- Queries have to execute in user defined queues.
- Runs in own space separate from WLM queues
- SELECT and CTAS queries only.
- Query runtime
 - Dynamic
 - User defined (1s to 20s)
- If runs more than SQA time then it is moved to WLM queue

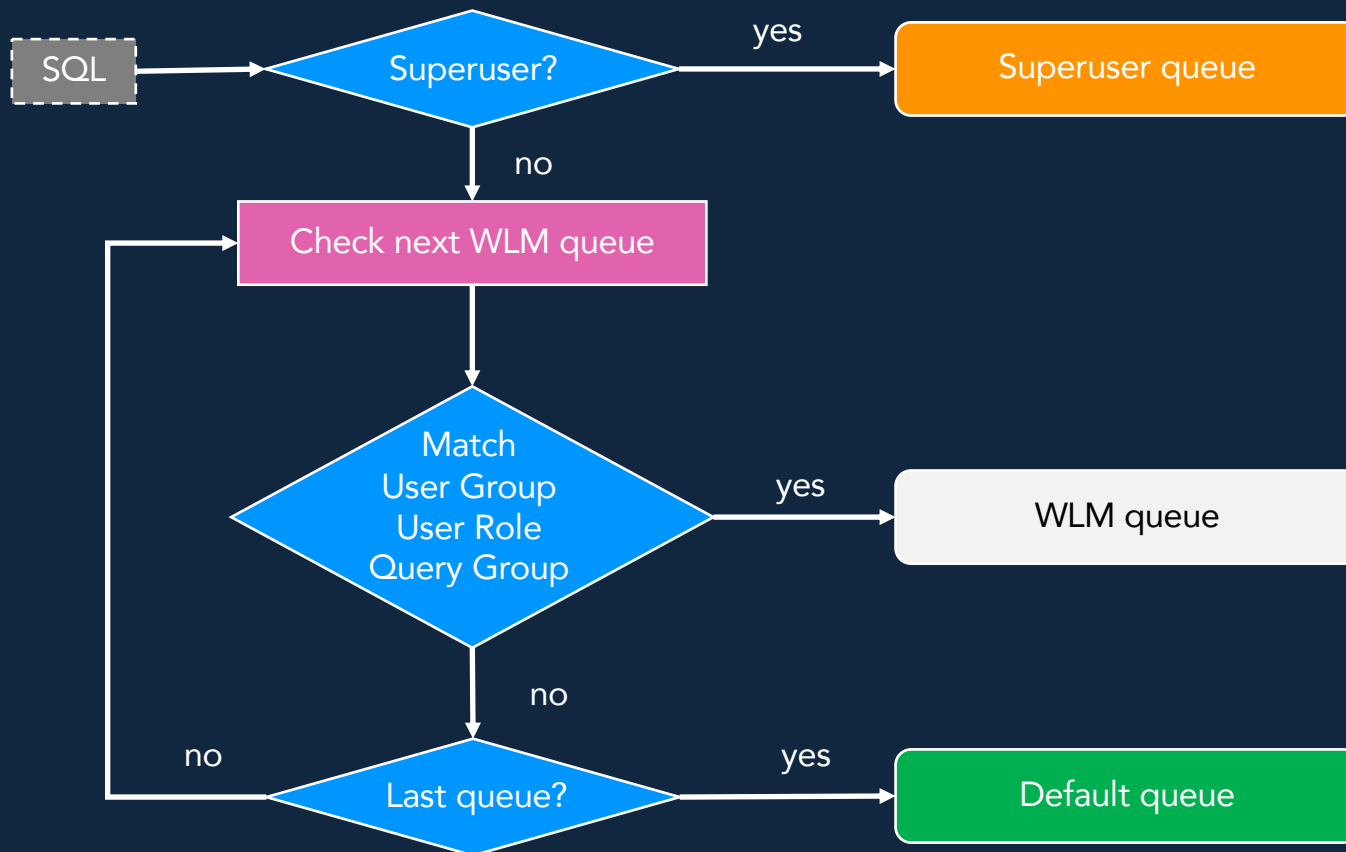


Query Assignment

- How queries are assigned to queues?
- Based on user groups.
- Based on user roles.
- Based on query groups at runtime.



Query Assignment



Query Monitoring Rules

- Define performance boundaries for queries based on metrics for user defined WLM queues.
- Available metrics
 - query_cpu_time – Only CPU time used by the query.
 - query_execution_time – Elapsed time for a query.
 - query_queue_time – Time spent waiting in the queue.
 - query_blocks_read – No. of 1 MB data blocks read by the query.
 - scan_row_count – No. of rows scanned in a step before filtering.
 - io_skew – Ratio of maximum blocks read by a slice to average blocks read by all slices.
 - return_row_count – Total no. of rows returned by the query.
 - query_priority – Priority of the query.

Query Monitoring Rules

- Define performance boundaries for queries based on metrics for user defined WLM queues.
- Available metrics
 - spectrum_scan_row_count – No. of rows in S3 scanned by Spectrum layer.
 - spectrum_scan_size_mb – Size of data in S3 scanned by Spectrum layer.
 - join_row_count – No. of rows processed in a join.
 - nested_loop_join_row_count – No. of rows processed in an NL join.
 - cpu_skew - Ratio of maximum CPU used by a slice to average CPU used by all slices.

Automatic WLM

- Redshift manages concurrency and memory allocation of the cluster.
- Max of 8 queues can be created.
- Concurrency is automatically adjusted based on workload and resource requirement of queries.
- Running complex queries with high resource requirement will reduce concurrency.
- Running queries with low resource requirements increase concurrency.
- Query priority
 - HIGHEST → HIGH → NORMAL → LOW → LOWEST.

Manual WLM

- We define all aspects of a query queue.
- Max 8 user queues can be defined.
- Concurrency scaling mode.
- Concurrency level or Query slots.
- User groups to queue.
- Query groups to queue.
- Memory percent to use.
- Query monitoring rules.
- Query queue hopping – Allows queries to temporarily jump to a higher priority queue if resources are available.

WLM System Tables

Table Name	Description
STL_WLM_ERROR	Contains a log of WLM-related error events.
STL_WLM_QUERY	Lists queries that are being tracked by WLM.
STV_WLM_CLASSIFICATION_CONFIG	Shows the current classification rules for WLM.
STV_WLM_QUERY_QUEUE_STATE	Records the current state of the query queues.
STV_WLM_QUERY_STATE	Provides a snapshot of the current state of queries that are being tracked by WLM.
STV_WLM_QUERY_TASK_STATE	Contains the current state of query tasks.
STV_WLM_SERVICE_CLASS_CONFIG	Records the service class configurations for WLM.
STV_WLM_SERVICE_CLASS_STATE	Contains the current state of the service classes.
STL_WLM_RULE_ACTION	Records details about actions resulting from WLM query monitoring rules associated with user-defined queues.
STV_WLM_QMR_CONFIG	Records the configuration for WLM query monitoring rules (QMR).

Hands-on

- WLM Parameter Group – Parameters.
- WLM Parameter Group - Workload management.
- Explore Manual WLM options.
- Explore Automatic WLM options.
- Create group "dashboard".
- Create group "reporting".
- Create users "u_rpt", "u_dsc", "u_mle", "u_dana", "u_dsh", "u_adm".
- Add "u_dsh", "u_dana", "u_mle" to "dashboard" role.
- Add "u_rpt", "u_dsc", "u_mle" to "reporting" group.
- Add to "u_adm" to "dba" group.

Hands-on – Manual WLM

- Enable SQA.
- Create 3 queues (service class) – rpt_queue, dsh_queue, admin_queue.
 - rpt_queue – 50% memory, CL 4, CS enabled, 60 mins timeout.
 - dsh_queue – 15% memory, CL 10, CS enabled, 30 mins timeout.
 - admin_queue – 30% memory, CL 2, CS enabled, 60s timeout.
- Check details for the queues defined above in [STV_WLM_SERVICE_CLASS_CONFIG](#).
- Run concurrent queries using Query Editor V2.
- Check concurrent queries in the cluster using the following and console.
 - [STV_WLM_QUERY_QUEUE_STATE](#)
 - [STV_WLM_QUERY_STATE](#)
- Check query queue summary in [STV_WLM_SERVICE_CLASS_STATE](#).
- Create Query Monitoring Rules (QMR) and check in [STV_WLM_QMR_CONFIG](#).

Hands-on – Automatic WLM

- Same operations as that of manual WLM.

Summary

- What is WLM
- Automatic and Manual WLM
- Query Queue
- Concurrency Scaling
- Short Query Acceleration (SQA)
- How queries are assigned to queues
- Query Monitoring Rules

